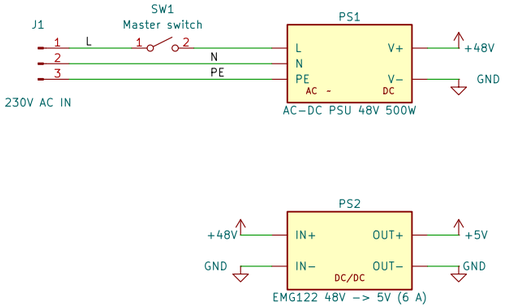
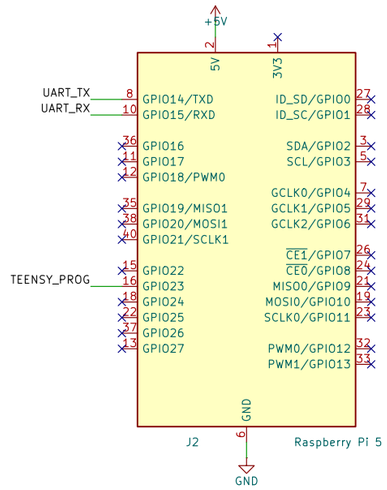
POWER SUPPLY

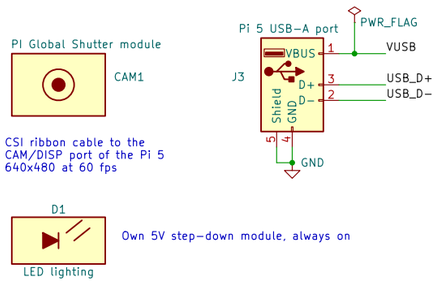


230V AC mains: L switched by SW1 (master switch).
PS1: 48V DC for the four DM556 power stages and PS2.
PS2: EMG122 step-down (10–90V in), 5V for the Raspberry Pi 5, max. 6 A.
SW1: 230V AC, 300 A. One common ground: PS2 is non-isolated.

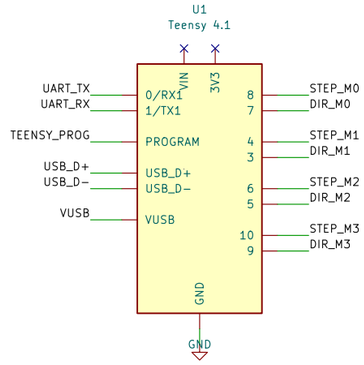
VISION – RASPBERRY PI 5



UART: /dev/ttyAMA0, 115200 baud, 3.3V.
Supplied through pins 2/4 (no USB-C PD):
USB ports limited to 600 mA.

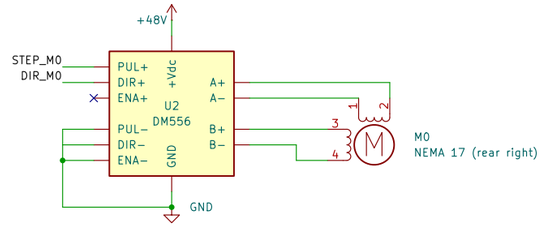


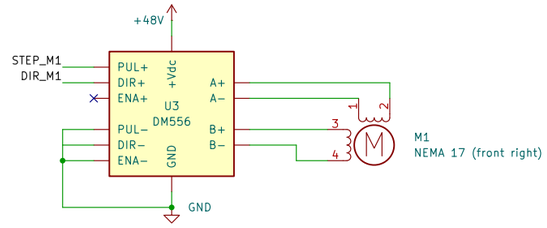
CONTROL – TEENSY 4.1

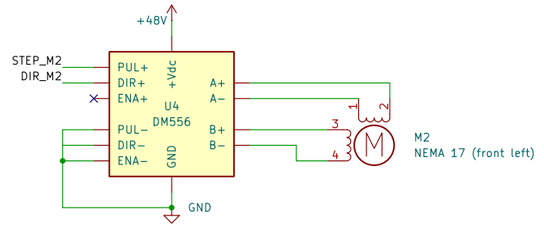


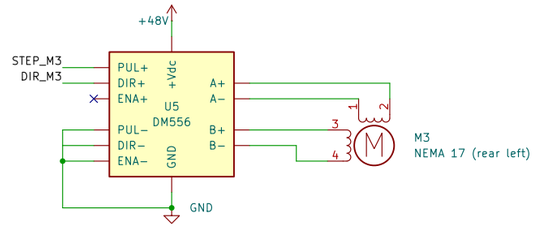
Supplied only through USB (VUSB)
from the Raspberry Pi 5.
200 Hz control loop, 3.3V outputs.
Encoder pins defined in the firmware
(CS 0, CAL 12, A/B 10/11) are unused.

MOTOR DRIVE









Common-cathode inputs: PUL+/DIR+ from the 3.3V Teensy outputs
(5V nominal); PUL–, DIR– and ENA– to the driver GND terminal;
ENA+ not connected (always enabled). 1/16 = 3200 pulses/rev.
DM556 current: 2.7 A RMS. Motors: Robofun NEMA 17, 1.5 A, 1.8 deg.

NOTES

Pin numbers: Raspberry Pi 5 = physical pins of the 40-pin header; Teensy 4.1 = digital pin numbers used in the firmware.
Nets with the same label are connected (UART_TX, UART_RX, TEENSY_PROG, USB_D+/–, VUSB, STEP_Mx, DIR_Mx).
UART_TX: PI GPIO14/TXD (pin 8) –> Teensy RX1 (pin 0). UART_RX: Teensy TX1 (pin 1) –> PI GPIO15/RXD (pin 10).
TEENSY_PROG: PI GPIO23 (pin 16) –> Teensy PROGRAM; puts the Teensy in bootloader mode for flashing.
Motor channels: M0 rear right (STEP 8, DIR 7), M1 front right (4, 3), M2 front left (6, 5), M3 rear left (10, 9).
Parts: PS2 EMG122 step-down (10–90V in, 6 A, non-isolated); camera: Raspberry Pi Global Shutter Camera with fisheye lens;
~~motors: Robofun NEMA 17 (1.5 A, 1.8 deg); DM556 set to 2.7 A RMS; SW1: 230V AC, 300 A; LED lighting: own 5V step-down, always on.~~